Entrepreneurship & Regional Development: An International Journal
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/tepn20

Entrepreneurs' networks and the success of start-ups

Peter Witt
[a] Otto Beisheim Graduate School of Management (WHU) , 56179 Vallendar, Germany E-mail:
Published online: 20 Feb 2007.

ENTREPRENEURSHIP & REGIONAL DEVELOPMENT, 16, SEPTEMBER (2004), 391–412

Routledge
Taylor & Francis Group

Entrepreneurs' networks and the success of start-ups

PETER WITT
Otto Beisheim Graduate School of Management (WHU), 56179 Vallendar, Germany; e-mail: pwitt@whu.edu

The network success hypothesis assumes a positive relation between the networking activities of founders and their start-up's success. The rationale behind this hypothesis is the theory of socially embedded ties that allow entrepreneurs to get resources cheaper than they could be obtained on markets and to secure resources that would not be available on markets at all, e.g. reputation, customer contacts, etc.

This paper clarifies how entrepreneurial network activities can be measured and which indicators exist to quantify start-up success. It then reviews empirical studies on the network success hypothesis. The studies have rarely come up with significant results. This surprising evidence can be explained by large differences in the way that the dependent and the independent variables were defined and by effects of unobserved variables such as the networking expertise of the founders and the entrepreneurs' level of existing know-how in the areas of co-operation and networking ('absorptive capacity'). The major shortcomings of existing network studies are found to be the neglect of different starting conditions, the focus on individual founders' networks instead of multiple networks in start-ups with an entrepreneurial team, and the assumption of a linear causal relation between networking and start-up success. Accordingly, the paper develops a new, extended model for the relation between entrepreneurial networks and start-up success. Finally, we make some suggestions for the further development of entrepreneurial network theory.

Keywords: networks; start-up success; co-operation; social embeddedness; costs of networking; network dynamics.

1. Introduction

Economic network theory has investigated a number of different exchange relations in networks and network actors. The typical object of observation is an individual person or an individual institution that has durable information contacts, exchange relations, or both, with other people or with organizations such as firms, universities, or authorities. Whenever the person or institution under survey has more than one contact of this kind that can be analysed, we are entitled to speak of a network and can conduct a network study.

Traditionally, network studies in the realm of strategic management and business administration theory have dealt with long-term relations between companies, i.e. co-operations, strategic alliances, joint ventures and others (Johanson and Mattsson 1987, McGee, Dowling and Megginson 1995, Witt 1999, Lechner 2001). In these studies, the network nodes are organizations (companies), the connecting lines are information or product exchange relations. Sociological approaches to network theory, which have a much longer tradition than economic ones, take individual persons as the nodes of the network and investigate communication or information links as the connecting lines between these persons (Bavelas 1948, Granovetter 1973, Freeman 1978/79).

Entrepreneurship and Regional Development ISSN 0898–5626 print/ISSN 1464–5114 online © 2004 Taylor & Francis Ltd
http://www.tandf.co.uk/journals
DOI: 10.1080/0898562042000188423

Both academic disciplines have found a fruitful intersection in entrepreneurship research where there is a long tradition of studying entrepreneurial networks and their effect on a start-up's success (Birley 1985, Aldrich and Zimmer 1986, Johannisson 1988). This line of research has come to be known as the 'network approach to entrepreneurship' (Brüderl and Preisendörfer 1998: 213). It is based on the hypothesis that founders use their personal network of private and business contacts to acquire resources and information that they would not (or not as cheaply) be able to acquire on markets. To put this in other words, the hypothesis is that entrepreneurs with larger and more diverse networks get more support from this network and thus are more successful than entrepreneurs with smaller networks or less support from their network.

This paper reviews the existing literature on the relationship between networking activities, the structure of entrepreneurial networks, as well as the services provided by network partners and start-up success. The paper critically examines empirical studies on the subject and develops a new, extended model for the relationship between entrepreneurs' networks and the success of start-ups. The aim is to set out a research agenda by formulation of a number of propositions that further empirical research should test. The paper also tries to give some suggestions for further theoretical research, in particular in the field of dynamic network theory, i.e. the development of entrepreneurial networks over time (Johannisson 1996, Hoang and Antoncic 2003) as well as the integration of sociological and economic network theories.

2. The relation between entrepreneurs' networks and the success of start-ups

2.1 Entrepreneurial networks

A network consists of single nodes (actors) and connections between these nodes (dyads), which as a whole form the structure of a network (Walker 1988). First, in this work we will have a closer look at the networks of founders as individual persons. A possible enlargement of the perspective that we will discuss later is to include networks of start-up teams and organizations. The relations of a founder under survey are mainly exchange relations for information and services. The potential network partners are other individual persons, e.g. family members, friends, business partners, other founders, but also contact persons at institutions such as universities, large companies, and authorities. The basis of the analysis is the founder and her relations to other persons, i.e. the focus will initially be on a so-called self-centred network. Extending this perspective, all relations among the founder's network partners will be analysed to create a truly networked point of view.

Many of the academic contributions to be reviewed in this paper have (explicitly or implicitly) assumed that the focal entrepreneur is a given, and moreover, homogenous individual whereas the network partners may be heterogeneous and not necessarily entrepreneurs. While it seems appropriate to focus on the networks of one entrepreneur (or better, as will be laid out later in this paper, the entrepreneurial team) and allow network partners to be all kinds of people (including other entrepreneurs), entrepreneurial characteristics and intentions obviously matter when investigating the relationship between networking activities and start-up success. Different types of entrepreneurs have different aspirations (Chell and Baines 2000) and different marketing capabilities (Smith 1967). Some founders have no 'growth willingness' (Davidsson 1989). Gender has also been found to be an important factor for start-up success (Chell and Baines 1998). We will get back to the problem of heterogeneous entrepreneurs, their intentions, and their networks when we investigate the independent and the dependent variable of the relationship between entrepreneurial networks and start-up success.

To describe a network's actors and structure, sociological theories have developed various quantitative measures. The whole network may be characterized by attributes such as density (Niemeijer 1973), i.e. the number of connections between the partners in relation to the number of maximum possible connections, connectedness (Bavelas 1948), or the extent of cluster formation. Network analysis describes the bilateral or dyadic ties between two persons within a network, i.e. the lines, by attributes such as symmetry, reciprocity, multiplicity (Lincoln 1982), and strength. The description of actors in networks has traditionally been dominated by the concept of centrality (Bavelas 1948, Nieminen 1974, Freeman 1978/79). A central person, e.g. within an information network, has many direct connections to other persons ('connectedness'), can reach other members of the network quickly, i.e. needs to use few or no intermediate persons ('closeness'), or is located on the information paths between other persons of the network frequently ('betweenness'). One limitation of all network research arises from the fact that empirical studies must use quantitative measures to estimate information which is essentially qualitative and cumulative in nature. The problem refers to data collection as well as data evaluation (Daft and Lengel 1986).

In economic and management research, the network perspective is considerably younger than in sociology. Nonetheless, there are examples of studies that have applied sociological centrality measures to the analysis of economic networks. They investigate the networks created by co-operative inter-firm relationships (Walker 1988), the networks of large corporations formed by interlocking directorates of their board members (Albach and Kless 1982), the supplier and customer networks of Eastern European firms in transition (Albach 1994), or the information networks of small and medium-sized enterprises (Witt 1999).

Granovetter (1985) has put forward the hypothesis that many economic transactions between persons are embedded into social relations and strongly influenced by them. This theory of 'social embeddedness' distinguishes typical market transactions without personal emotions between the transaction partners (so-called 'arm's length relations') and transactions embedded into permanent social relations (so-called 'embedded ties'). The new insight of the theory of social embeddedness is that only the arm's length relations are really handled like standard economic theory predicts, i.e. guided by short-term, selfish, and profit-maximizing behaviour of the people involved. The transaction partners of embedded ties trust each other, show reciprocal instead of profit-maximizing behaviour and take a long-term perspective on the relation. An empirical study by Uzzi (1997) shows that socially embedded transactions occur less frequently than market transactions, that they are the preferred mode of interaction for especially important exchanges, and that they generate higher benefits for all participants than arm's length transactions.

2.2 *Network theories in entrepreneurship research*

Entrepreneurs build organizations that enable them to seize market opportunities (Larson and Starr 1993). In doing so, they compete with well-established companies and face at least two disadvantages: the small size of their firms in early stages of the development process (liability of smallness) and their companies' lack of reputation and corporate history (liability of newness). Therefore, entrepreneurship research has traditionally been trying to give explanations of why at least some start-ups prosper and grow in competitive environments.

One prominent explanation for start-ups' success has explicitly referred to network theory and investigated the personal networks of entrepreneurs and their effect on start-up performance (Birley 1985, Aldrich, Rosen and Woodward 1987, Johannisson 1988). We will call this line of research the 'network success hypothesis' of entrepreneurship theory, using a term coined by Brüderl and Preisendörfer (1998: 213). The network success hypothesis states that founders can gain access to resources more cheaply by using their network contacts than by using market transactions, and that they can even acquire resources from the network that would not be available via market transaction at all. 'In summary, entrepreneurs can increase their span of action through their personal networks and gain access at a limited cost to resources otherwise unavailable' (Dubini and Aldrich 1991: 308).

It is a general finding that the utilization of resources stemming from network contacts may offer advantages in comparison to the reliance on resources acquired via market mechanisms, i.e. arm's length transactions. This has proven to be true for large firms as well. Jarillo (1989) empirically investigated 1902 stock-listed corporations in the USA and found that firms making intensive use of network resources (external resources) grow significantly stronger over a 10-year observation period than firms focusing on internal resources, i.e. resources that the company owns. The major shortcoming of Jarillo's (1989) analysis is that he measures the role that external resources play in a firm's production process by the ratios of sales to balance sheet total and sales to total number of employees. This is a rather crude measure for the degree of utilization of network contacts and gives no indication about the precise nature of the advantages that the network offers over external purchases of resources.

The opportunity to procure resources for a start-up at favourable rates from the personal network arises due to friendship or kinship ties to network partners. They offer the entrepreneur specific resources at no charge or below the market price simply to do them a favour or to return a favour that they received earlier. Practical examples are spouses working in start-ups without a salary, the provision of new or used production equipment by business friends for free, or the bargain price of a friend who is a tax consultant and helps the entrepreneur with accounting problems. The cheaper and the more frequently resources are available in an entrepreneur's personal network, the better her chances are to realize cost advantages over competitors (Starr and MacMillan 1990: 83–85). Examples for resources obtained from the network that would not be available for a start-up on the market are reputation, e.g. from an experienced manager serving on the start-up's supervisory board, orders from large corporations that would normally not order from start-ups, e.g. because a former colleague and friend is in charge of procurement at the large corporation, or new, proprietary, and difficult to purchase (sticky) information on technologies (Hippel 1994), e.g. coming from a former thesis advisor at a university.

2.3 *Measures for entrepreneurial networks*

To be able to test the network success hypothesis empirically, we need to define the independent variable very precisely, i.e. lay out the relevant network concept and propose measures for it. In the causal chain between an entrepreneur's network and her start-up's success, one needs to distinguish between three different levels of observation. First, the analysis could focus on the creation of entrepreneurial networks, i.e. the activities that an individual entrepreneur undertakes to build, sustain, or extend her personal network. This version of the independent variable will be called 'networking activities'. Second, the analysis could be directed at the structural characteristics of an entrepreneur's network at a certain point in time, which is equivalent to measuring the result of earlier networking activities. We will call this alternative to define the independent variable of the network success hypothesis 'network structure'. Third, and theoretically closest to firm performance, one could measure the economic benefit of the information and the services received from network partners over a certain period of time. We will call this version of the independent variable which can (and should) include the economic costs of maintaining and utilizing a personal network, the 'benefits received from the network'.

Obviously, these alternative proxies are interdependent. The more networking activities an entrepreneur engages in, the larger her personal network and the more central her position in it should be. The more favourable we judge an entrepreneur's personal network structure, the more benefits we expect her to receive from this network. On the other hand, some founders have no aspirations to be successful in the sense of creating growing companies, so they may deliberately restrict their network size (Chell and Baines 2000). Founders have heterogeneous networking abilities, so their 'absoprbtive capacity' (Cohen and Levinthal 1990) to derive benefits from existing network partners may vary. Figure 1 shows the relation we have assumed here between entrepreneurial networking activities, network structure, and

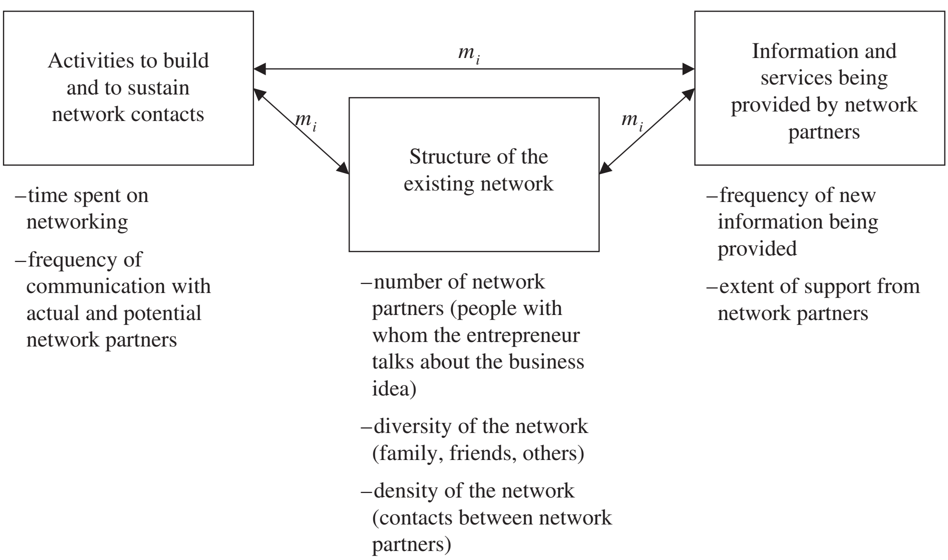

Figure 1. Measures for entrepreneurial networks.

benefits received from the network. The moderating variables m_i are specific for each start-up i and indicate that the causal links between the different versions of the independent variable depend on external factors, for example the founders' entrepreneurial aspirations, networking abilities, 'absorptive capacities', and gender.

To measure individual networking activities empirically, a number of different items have been suggested. One concrete proposal is to ascertain the amount of time an entrepreneur invests per month in the creation, the preservation, and the enlargement of her personal network. A similar measure is the number of hours spent per week by the entrepreneur to acquire new business contacts and to maintain existing ties (Aldrich and Reese 1993). Another suggestion is to investigate the frequency of communication between the entrepreneur and (actual and potential) network partners per week (Ostgaard and Birley 1996).

Measuring structural network properties is the second and certainly also the most frequently chosen way of entrepreneurship research to make the network success hypothesis operational. A first structural measure is the size of an entrepreneur's personal network, i.e. the number of different persons with whom the founder has talked about her business plan or the business idea (Aldrich, Rosen and Woodward 1987, Aldrich and Reese 1993). Another structural measure is the network diversity, i.e. the heterogeneity of network participants. One idea is to classify network partners into three groups, family, friends, and acquaintances, and then to measure the number of people in each group. The theory behind this procedure is Granovetter's (1973) model of strong and weak ties. This model defines strong social ties as relations with high levels of emotional underpinning, e.g. relations to family members and friends. Weak ties can be utilized without these emotional components and are based on more rationally dominated relations, e.g. to colleagues, bosses, business partners and the like. Granovetter's famous hypothesis is the 'strength of weak ties'. This hypothesis postulates strong ties to be very reliable but also characterized by a large degree of redundancy in terms of the information being exchanged. Weak ties are less reliable but offer better access to new information. With respect to entrepreneurship, a personal network structure with a balanced mixture of strong and weak ties, i.e. a heterogeneous network, has been regarded as especially favourable to the founder's economic success (Uzzi 1997, 1999). Another structural measure is the network's connectedness which describes the number of direct relations between the entrepreneur's personal network partners, i.e. the density of the network (Hansen 1995).

The third group of network measures directly targets the benefits obtained from entrepreneurial networking activities, respectively, the structural properties of their personal networks. Examples are attempts to quantify the number and the value of network services that entrepreneurs received via network contacts (Brüderl and Preisendörfer 1998). Such network measures have the advantage to be closer to the start-up's economic success than the other two groups. Their main disadvantage is that they leave an important question of entrepreneurial network theory unanswered, namely what entrepreneurs can do to improve their chances of success in terms of networking and influencing the structural characteristics of their personal network.

This survey of alternative ways to measure the independent variable in the network success hypothesis indicates that empirical studies using different concepts for the entrepreneurial network may come to very different results simply because the independent variable has been given very dissimilar interpretations, or, as Salancik (1995: 355) formulated: 'There is a danger in network analysis of not seeing the trees for the forest. Interactions, the building blocks of networks, are too easily taken as given'.

2.4 *Measures of the success of start-ups*

To test the network success hypothesis empirically, the dependent variable, i.e. a start-up's performance, needs to be clearly defined and suitably measured as well. Depending on the company's state of development in the foundation process, there are very different possibilities to define success (Brush and Vanderwerf 1992, Chandler and Hanks 1993), which will be presented briefly below.

A first suggestion for a success measure is the completion of the idea and planning phase, i.e. the founder has moved from idea development and business planning to business start-up. This criterion relates to the entrepreneur and not to the start-up company. The fact that she/he has been able to move to this next stage may be considered to be a success, although it is not an overly restrictive success measure. Completing the idea and planning phase suggests more about entrepreneurial intention and commitment than it says about start-up success.

A second success measure that also relates more to the entrepreneur and less to the start-up is the subjective evaluation of entrepreneurial success by the founder. The crucial disadvantage of this approach is the fact that founders may have very different expectations about their life as an entrepreneur. Thus, the subjective satisfaction of a founder is dependent on objective success criteria such as salary, increases in company value, and the like, but also on individual expectations and feelings: 'There is reason to believe that different people may not be equally satisfied with the same level of performance, and thus reason to doubt that a satisfaction with performance index provides a good proxy for firm performance' (Chandler and Hanks 1993).

A prominent, non-subjective, i.e. company-related success measure is the survival of a start-up company, its persistence in the market. Data on this success measure can be obtained comparatively easily. Given the date of foundation of a start-up in a sample of actually founded companies, researchers can investigate for each sample firm if it is still existing at the time of the enquiry. This can be done by calling the company, visiting it personally, or visiting its web site. To eliminate biases owing to diverse survival periods, the sample could consist of only those start-ups that were founded in the same year. Cross-sectional analyses and panel analyses on the basis of repeated enquiries of a set of actually founded companies show how long individual firms have survived and which start-ups left the market after what time periods. The most important methodological problem with survival as a measure of the success of start-ups is the determination of a suitable period of time after which survival is to be stated. If this period is too short, the success measure is not demanding enough. Survival in the short run may simply be due to high initial levels of capital in combination with low cash burn rates. If researchers choose too long a reference period, the focus shifts from start-ups to established companies.

A second group of measures for the success of start-ups refers to a company's growth rates. Typical growth indicators are sales, the number of employees, or the balance sheet total. This type of data can best be obtained in empirical studies using personal interviews or questionnaires. When searching for suitable ways to make growth operational, the problem of absolute and relative growth arises. The utilization of annual growth rates creates the bias that small firms will be classified more easily as successful than large firms. For absolute increases in sales and employee figures, the opposite bias holds. To reduce the company size effect on the success measure as far as possible, the use growth indices are recommended that consist of absolute and relative growth measures. Another methodological problem of growth as a success measure for start-ups relates to the viewing period. Some empirical studies calculate a 3-year compounded annual rate of sales growth (McGee, Dowling and Megginson 1995: 569). Other authors have used a growth rate based on the ratio of the current number of employees and that at the time of foundation. If the companies in the sample are of different age, this procedure obviously produces large biases in terms of the calculated growth rate. The older a start-up is, the better its chances are to have realized a large growth rate.

A third group of company-related success measures aims at later stages of a start-up's development process and calculates ratios, which are common in the analysis of large and well-established companies. Typical examples are profits and return on investment. The main problem here is the trade-off between growth and profitability. Some start-ups pursue business models that explicitly forego profits in order to realize large growth rates. 'Profitability and growth measure different aspects of performance as growth is sometimes achieved at the expense of profitability in the short run' (Lee and Tsang 2001: 586). Another problem is more frequent in practice: in personal interviews or in answering questionnaires, entrepreneurs are frequently unwilling to disclose information on financial performance indicators (Brush and Vanderwerf 1992). Finally, the methodological problem of selecting a suitable viewing period occurs again.

A general problem of all objective measures for start-up success is that they depend on the founders' intentions and aspirations. An empirical study of 400 Swedish small business owners from 1989 has shown that significant relations exist between expected outcomes and growth willingness. In 40% of the sample firms, there was no intention to grow at all, due to fears of reduced employee well-being and a loss of supervisory control (Davidsson 1989). A UK study of small service firms from 1998 found that gender affects business performance (Chell and Baines 1998). Similarly, 'entrepreneurs' may have more ambitions to make profits than 'craftsmen' (Smith 1967). What follows from these studies is that, in a very general setting, intentions, gender, marketing capabilities, etc., should be control variables if start-up success is being measured by objective criteria such as profitability, growth or increases in firm value.

This paper will not delve into the problem of entrepreneurial characteristics and intentions any deeper but focus on entrepreneurs willing to grow their businesses, to make profits, and to realize increases in value. Perhaps the easiest way to justify this assumption is to restrict the analysis to high-tech companies being financed by venture capital. They need to grow quickly to reach minimum efficient scale, they have to realize increases in corporate value to make exits possible for the venture capital firm, and they are interested in profits (not so much or not only in independence, craftsmanship, or technology) because otherwise they would not have teamed up with a venture capital firm (Sahlmann 1990).

If we summarize our arguments, this section has shown the following: in selecting an appropriate criterion to measure the dependent variable of the network success hypothesis, company-related measures are to be preferred over subjective, personal evaluations of start-up success. As a general rule, the measure for start-up success should be chosen depending on the stage of development of the venture. Figure 2

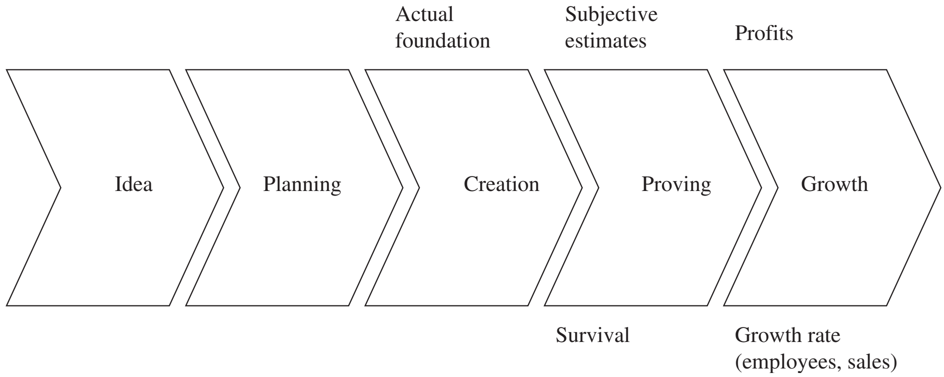

Figure 2. Measures for start-up success.

summarizes the possible measures for start-up success that we suggested for each development phase.

2.5 *Empirical tests of the network success hypothesis*

Aldrich, Rosen and Woodward (1987) conducted one of the first empirical studies that looked at the effect of entrepreneurial networking activities on the success of the respective start-up. The authors interviewed 285 potential and actual founders in North Carolina in 1986. Ten months later, 212 participants of the first round were questioned again. In total, 165 responses from both rounds could be analysed statistically. Aldrich, Rosen and Woodward (1987) measure entrepreneurial networks, the independent variable, with three indicators: network size, network diversity, and the ease of access to network resources. The dependent variable, i.e. start-up's success, is determined as the decision of a potential entrepreneur to realize the business idea and the start-up's profitability. The main findings of the empirical study follow.

The accessibility of network resources is significantly and positively correlated with the decision to start a new business. In start-ups that are less than 3 years old, network diversity correlates negatively and the accessibility of network resources positively with the start-up's profitability. For firms older than 3 years, the authors find a positive relation between the size of the entrepreneur's network and the company's profitability. It is noteworthy that the original study by Aldrich, Rosen and Woodward (1987) has been replicated in a number of countries such as Italy (Aldrich *et al*. 1989), Sweden (Johannisson and Nilsson 1989), Northern Ireland (Birley, Cromie and Myers 1991), and most recently in Greece (Drakopoulou Dodd and Patra 2002). Comparing the results of this series of linked, but not methodologically identical studies, Drakopoulou Dodd and Patra (2002: 119) find 'some homogeneity, suggesting a degree of generic universal entrepreneurial behaviour, and some heterogeneity, highlighting the importance of cultural differences'.

Cooper, Folta and Woo (1991) analyse a sample of 2246 companies that were established between 1984 and 1985 in the USA. The founders of these companies answered questionnaires twice, for the first time in 1985 and for the second time in 1986 and 1987. The authors distinguish between three different sources of information for entrepreneurs: public sources (books, associations, etc.), personal sources (friends, relatives, acquaintances), and professional sources (bankers, tax consultants, lawyers, etc.). The independent variable is networking activities which is measured as the intensity of using personal sources of information. The dependent variable, i.e. start-up success, is survival after 2 or 3 years. The study does not find any significant correlations between the two variables.

Aldrich and Reese (1993) analyse entrepreneurs from North Carolina to whom they sent questionnaires twice, first between 1990 and 1991 (444 usable answers) and, second, via telephone interviews 2 years later (281 usable answers). The authors characterize the networking activities of the entrepreneurs in their sample with the following variables: network size as well as time spent developing and maintaining business contacts. The study classifies survival (continuation of the same business under the same owner) after 2 years and increases in revenue as entrepreneurial success but finds no correlation to the networking variables.

Hansen (1995) tests for pre-founding social structure and process effects on subsequent first-year new organization growth rates. The sample of this study consists of 44 entrepreneurs from Tennessee, USA, who had founded new businesses. The three criteria to measure networking activities were: the size of the subset of people from the entrepreneur's network who had been involved in founding the new organization (active network); the density of the entrepreneur's network; and the frequency of communication within the network. As a proxy for start-up success, Hansen (1995) selects a somewhat uncommon measure, the first-year growth of the start-up in terms of the dollar amount of the start-up's monthly payroll. This measure intends to not only capture increases in the number of employees but also increases in qualification (salary) of existing employees. Using a multivariate regression analysis, the study finds significantly positive correlations between the size of the active network subset and the respective start-up's success. The same is true for the density of the entrepreneurial network.

Johannisson (1996) was the first to test the network success hypothesis for a European sample of entrepreneurs. He analysed 361 potential and actual founders in Sweden in 1987 and replicated the study in 1993 with 158 persons from the same sample. Johannisson (1996) collects two groups of network data. The first group consists of measures for the whole network of an individual entrepreneur, in particular network size and the monthly investment (in hours) to maintain and enlarge the network. The second group focuses on a subset of the whole network, called the 'primary network', i.e. the five people the entrepreneur prefers to talk with about her business. For this group, the study evaluates four measures of the independent variable networking: the character of relationship (business/social), the frequency of exchange (daily, weekly, less frequent), the strength of ties between the different subset network participants (well acquainted, acquainted, strangers, unknown), and the size of each primary network partner's own personal network (very large, large, small, unknown). A start-up is classified as successful if the entrepreneur herself thinks she is successful in terms of growth prospects, financial performance, and personal fulfilment. Overall, the study does not find a significant correlation between any of the independent and dependent variables.

Ostgaard and Birley (1996) also test the network success hypothesis for Europe. They create a sample of 159 entrepreneurs in the two English counties of Cambridgeshire and Avon to whom they sent questionnaires in 1991. The authors use a variety of measures for entrepreneurial networks and entrepreneurial networking, which they summarize to four groups of variables: network size (number of network partners); time spent to maintain and enlarge the network; network diversity;

and the intensity of using the network. Three different growth rates over 3 years are taken as proxies of start-up success, the growth rate of sales, the growth rate of employment, and the growth rate of profits. Univariate and multivariate regression analysis leads to two major conclusions: the size of the entrepreneur's network and the time spent to maintain and enlarge the network both show a positive and significant correlation with the growth rate of employment (but not with the other two success measures).

In a German study to empirically test the network success hypothesis, Brüderl and Preisendörfer (1998) investigate a large sample of start-up firms ($n = 1710$) that were founded between 1985 and 1986 in Munich and Bavaria. Deviating from the preceding literature, the study does not use networking activities or structural properties of an entrepreneur's network as the independent variable, but the degree of support that an entrepreneur receives from her personal network. The authors use three measures for start-up success: survival 4 to 5 years after foundation; the growth of the number of employees; and the growth of sales. The core result of the work by Brüderl and Preisendörfer (1998) is that a large degree of support for entrepreneurs from their personal networks significantly increases the chances for survival and growing sales. In particular, emotional support from the family has a significantly positive effect on start-ups.

Littunen (2000) investigates 129 start-ups from Finland of which 110 were continuing firms and 18 were closed firms. The author chooses survival beyond the critical operational phase (4 to 6 years) as the criterion for success. Two dichotomous variables measure networking activities, i.e. the independent variable of the network success hypothesis: co-operation between firms in the start-up phase and changes (increases and decreases in the number of partners) in the external personal networks of entrepreneurs. Using a logistic regression model, the study finds no significant correlations between networking and start-up success.

3. Shortcomings of existing theoretical and empirical studies

3.1 The problem of different starting conditions

In its most general form, the network success hypothesis postulates that building and maintaining large personal networks is recommended for entrepreneurs to foster the success of their start-ups. However, there are some theoretical insights and even empirical findings from other areas of entrepreneurship and management research that make the argument in its general form rather doubtful.

Utilizing network contacts in ways that make them beneficial for the development of a start-up requires know-how and co-operation qualifications on the side of the entrepreneur that may not be present in all cases. McGee, Dowling and Megginson (1995) have shown empirically that using external resources by co-operating with other firms will only increase corporate success if the management team of the firm under consideration has extensive know-how in the area of co-operation.[1] Cohen and Levinthal (1990) have introduced the concept of 'absorptive capacity', which depicts the same idea: founders will not be able to benefit from co-operations with and information from network partners if they do not possess the necessary knowledge and the capacity to absorb the information in their own organization. Therefore, entrepreneurial experience, social competencies, the level of university education and other factors determine if and how much benefit an entrepreneur can derive from existing network ties. 'Thus neither the frequency nor the regularity of exchanges necessarily indicate the potency and reliability of the ties' (Johannisson 1988: 85).[2]

A second restriction on the general recommendation derived above from the network success hypothesis is the fact that entrepreneurs may differ largely in terms of available own resources such as financial capital, know-how, patents, etc. Bayer (1991) has criticized the network success hypothesis accordingly and put forward a completely different 'compensation hypothesis'. It states that only entrepreneurs who are ill-equipped with resources build large networks and intensely seek for support from network partners. If the amount of the entrepreneur's own resources at the foundation date has a direct and positive influence on the chances for survival, then entrepreneurial networking activities merely compensate for comparably worse starting conditions. In a similar line of reasoning, Chicha (1980) has shown empirically that French small and medium-sized businesses increasingly utilize network contacts the more economic problems they encounter. The hypotheses of Bayer and Chicha (1980) have direct implications for empirical research: it would be impossible to find positive correlations between strong networking activities and a start-up's success even if networking as such was beneficial.

Finally, the size, the strategy, and the industry of a start-up can have a moderating effect on the correlation between entrepreneurs' networks (or networking activities) and their companies' success. Start-ups do largely vary in size even if they are in the same stage of the development process, but most empirical studies do not control for the size effect. Lee and Tsang (2001) have tested a causal model of entrepreneurial success for a sample of Chinese entrepreneurs in Singapore. The authors show that the size of an entrepreneur's communication network correlates positively with the growth rates of the respective start-up but that this effect is much stronger for large than for small firms. Kirchhoff (1994) notes that start-ups in some 'glamorous' industries such as biotechnology need larger and more international networks than others. Burt (1992) has developed the theory of 'structural holes', which relates networking activities to the competitive strategies that companies pursue. Thus, future empirical studies on the network success hypothesis need to take into consideration that it may depend on the start-up's resource base, size, strategy, and industry, as well as, for example, the gender of the founders.

3.2 The problem of multiple networks

All existing empirical studies focus on the evaluation of one individual entrepreneur's network. Unfortunately, in quickly growing start-ups we find teams of entrepreneurs much more frequently than single entrepreneurs. Every team member contributes resources and information, part of which has been obtained via the network ties. Therefore, empirical tests of the network success hypothesis should be related to the personal networks of all members of the founding team even if that leads to considerably more effort in data gathering.

If we try to take multiple networks of entrepreneurial teams into consideration, a number of theoretical questions emerge. Perhaps the most simple assumption is that the networks of the founders in a start-up are additive, i.e. the network size is the number of direct contacts of all founders combined. The problem with an additive model of team networks are overlaps in the personal networks of the individual team members. To give an example, if two founders are well acquainted with the same person, one of the network ties could simply be redundant. In this case, the appropriate assumption is that the number of direct relations to different persons defines the size of the team's network. However, there is also good reason to believe that such 'double' ties are stronger and more valuable than 'single' ties. To make an assumption as conservative as possible, it is suggested only to count those network ties of additional founders as ties of the team's network where no other team member has a network relation to the same partner already.

Extending the idea of multiple networks, employees and their personal networks may be considered as well. Bouwen and Steyaert (1990) have developed a model of how entrepreneurial teams build networks. It is rooted in psychological theory and explains network building and defining tasks for employees as two parallel and interdependent processes which constitute the 'organizing texture' of a start-up. By hiring new employees, the entrepreneurs extend their network and redefine their entrepreneurial tasks. In the terminology of this paper, whenever employees utilize their network contacts to acquire resources for the firm at prices below market value, they are directly contributing with their personal network to the company's success. There are no empirical studies on the importance of employees' networks for start-up performance yet. Theoretically, they can be less important than the entrepreneurs' networks (or even irrelevant), equally important, or even more important.

Given the lack of empirical results, there is only two obvious conclusions to be drawn from the theory of multiple personal networks in start-ups. First, whenever a new businesses is started by an entrepreneurial team and not by an individual entrepreneur, the analysis needs to address the networks of the team instead of one single member of the team. Second, future research needs to bridge the gap between current entrepreneurship theories focusing on individual networks and strategic management research that deals with theories of organizational networks (Larson 1992). As Johannisson (2000: 378) has pointed out, 'one reason for not seeing the relationship between networking activity and firm growth may be that an inappropriate unit of analysis has been adopted'. The more a start-up proceeds in its development and the larger it grows, the more relevant the personal networks of employees and external managers in the organization become. In the long run, corporate success will depend more on the network and the networking activities of the whole organization than that of an individual entrepreneur.

3.3 The problem of non-linearity

All empirical studies reviewed in this paper neglect the costs of networking activities.[3] However, building and maintaining network relations is obviously costly for entrepreneurs. Apart from the opportunity costs of investing time in networking activities, network ties are based on trust and on reciprocity. To be more precise, an entrepreneur cannot only ask network partners for information and access to cheap resources, she needs to contribute to the network as well. Although network relationships 'have no books recording the exchanges' (Johannisson 1988: 84), in the long run the exchanges between two partners need to be balanced. Game theoretical experiments have proven that most people are driven by reciprocity, i.e. the desire to be kind to those who have been kind to them (Fehr and Gächter 1998). The costs of networking activities for entrepreneurs stemming from reciprocity may not be linearly rising in network size (number of partners). At least for the opportunity cost of time, it is more plausible to assume that the marginal costs of adding a new network partner to one's personal network increase.

Only few other theoretical studies take the cost of networking into consideration when analysing entrepreneurs' or firms' networks. Johannisson (1996) as well as Ebers and Grandori (1997) have pointed at the importance of two types of costs, direct costs of delivering services or information to network partners and indirect costs in the form of the opportunity costs of time. None of the empirical studies which have been reviewed for this paper measures the costs of networking. Therefore, it is not surprising that they find no positive relationship between general features of the personal network of the founder, such as time invested and the scope of the network, and her venture performance (Johannisson 2000: 378).

Non-linearities may be found for benefits received from networks as well. All existing empirical studies have tested linear versions of the network success hypothesis with respect to support received from network ties. Only a few authors have laid out the theoretical argument that entrepreneurs can invest too much time in networking activities. Aldrich and Reese (1993) indicate the theoretical possibility of a non-linearity (but do not adapt the design of their empirical study accordingly): 'As in so many other things in life, the golden mean may lie somewhere in the middle' (Aldrich and Reese 1993: 334). Uzzi (1997) analyses the personal networks of entrepreneurs in the textile industry in New York. He finds that too large an extension of social networks reduces the flow of new information to the entrepreneur and causes inefficiencies. Uzzi (1997: 58) uses the term 'overembeddedness' to depict networks which are too large and which he was able to identify in a later study of entrepreneurial networks as well (Uzzi 1999).

Witt (1999) has developed a model to evaluate the position of small and medium-sized firms in their respective information networks. In his model, the potentially non-linear benefits of individual network ties depend on the frequency of utilizing a tie and the average value of the information being exchanged. The latter variable depends on the degree of trust between the partners, the relevance of the information for the firm, the speed of information transfer, the hierarchical position of the information partner, and his centrality in the overall network. Unfortunately, Witt's valuation model (1999) does not include the costs of networking.

In a related theoretical study, Witt and Rosenkranz (2002) have investigated entrepreneurial networks and proposed a model for the valuation of individual network ties and overall networks that includes non-linear benefits and costs of networking. On the level of individual ties between an entrepreneur and a network partner, the model defines the relevance of each partner for the start-up business, the partner's accessibility, and the cost of maintaining the tie to the partner for the entrepreneur as the core value drivers. There may be a direct trade-off between relevance and accessibility, e.g. if the entrepreneur knows a very influential manager of a large company but has only few opportunities to get in contact with that person.

4. An extended model of the relationship between entrepreneurs' networks and the success of start-ups

In its most extensive version, the network hypothesis postulates a (hopefully positive) relationship between entrepreneurial networking activities and start-up success. Before measuring entrepreneurial activities of individual founders or better – as suggested earlier – of entrepreneurial teams, the starting conditions of networking activities need to be controlled for. Entrepreneurship and strategic management research suggest that founders are not equally well prepared and willing to undertake networking activities. In some industries, e.g. biotechnology (Kirchhoff 1994), networking is more important than in others because implicit and tacit knowledge is necessary to keep competitive advantages. Firms that pursue strategies relying to a large extent on co-operation with other firms need more networking than start-ups with a lesser degree of co-operation. Founders who are ill-equipped with resources (Bayer 1991) or face economic difficulties (Chicha 1980) may feel a stronger pressure to pursue networking activities in order to get access to resources. Finally, men may be more used to building and utilizing networks than women (Aldrich *et al.* 1989) so that the willingness of the entrepreneurial team to invest time and money in networking activities may be larger when more men are members of the team. We summarize our arguments in the following proposition.

> *Proposition 1*: Founders invest more time and money in networking activities the more their industry is based on tacit knowledge, the more their strategy requires co-operating with partners, the less well equipped the start-up is with resources, and the more men are members of the founding team.

Following the results from existing network studies, we expect a positive relationship between the degree of entrepreneurial networking activities and structural measures of the founders' network. In other words, the more time and money the entrepreneurial team invests in the maintenance and the extension of the network, the larger and the more diverse it should be. One important caveat applies. More networking does not automatically produce larger or more diverse networks. It depends on the founders' networking abilities (Burt 1992, Baron and Markman 2000) and on national cultural settings (Drakopoulou Dodd and Patra 2002) if and to what extent networking activities lead to better measures of the team's network. The proposition reads:

> *Proposition 2*: There is a linearly positive relation between the networking activities of all founders and the structure (size and diversity) of their aggregated personal network, but the magnitude of this effect depends on the founders' networking abilities as well as specific conditions of national culture.

Given the above-mentioned arguments on the costs and benefits of network ties for entrepreneurs starting a new venture, the traditional network success hypothesis can be modified in its most frequently postulated form, i.e. the suggested positive relation between structural characteristics of the network and the net benefits received from it. Whether founders (the members of the entrepreneurial team) can really utilize their network to obtain valuable information and services depends on the firm's size (Lee and Tsang 2001), its absorptive capacity (Cohen and Levinthal 1990, McGee, Dowling and Meggnison 1995), and the reciprocity costs of utilizing personal relations for business purposes. Furthermore, as the marginal benefits of using the network are possibly decreasing and the marginal cost of doing so possibly increasing, the relation between network structure (size and diversity) and the respective start-up's success is expected to be inversely U-shaped, i.e. there is an optimal network size and diversity. We therefore derive the following proposition.

> *Proposition 3*: There is a non-linear (inversely U-shaped) relation between the structure (size and diversity) of the aggregated network of all founders and the net benefits they receive from network partners, but the magnitude of this effect depends on the 'absorptive

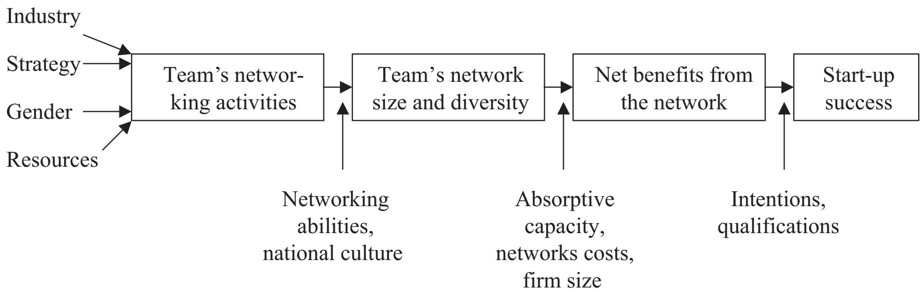

Figure 3. A new model for the relation between entrepreneurs' networks and the success of start-ups.

capacity' of the founders, the costs of obtaining information and services from network partners, and firm size.

The most narrow version of the network success hypothesis expects a positive relationship between the net benefits a founding team derives from the network and the success of the start-up. While this is certainly plausible and needs no modification as long as success is measured by subjective estimates of entrepreneurial success, the model needs to define moderating variables to control for different entrepreneurial intentions and qualifications if success is being measured by objective firm performance criteria. Therefore, we come up with two variants of a fourth proposition.

Proposition 4a: There is a linearly positive relation between the net benefits founders receive from their network partners and the success of their start-up in terms of their subjective estimation of entrepreneurial success.

Proposition 4b: There is a linearly positive relation between the net benefits founders receive from their network partners and the success of their start-up in terms of profitability, growth, and value creation, but the magnitude of this effect depends on the founders' intentions with respect to profitability, growth, and value creation as well as their qualifications.

Figure 3 summarizes the structure of this new model for the relationship between entrepreneurial networks and start-up success.

5. Suggestions for further research

5.1 Dynamic network models

The existing network theories of entrepreneurship and the new model proposed in this paper all implicitly assume that the actors, i.e. the entrepreneurs, can evaluate and influence their network ties to other persons. In particular, existing research is guided by the idea that personal networks of entrepreneurs are being created by economic decisions and not simply by chance. Most existing theories are also static in the sense that they derive no hypotheses about changes in network structure and its effect on entrepreneurial success over time.[4] A dynamic network theory pursues the goal to explain the development of entrepreneurial networks and their contributions to a start-up's performance over time. Hoang and Antoncic (2003: 175) use the term 'networks as the dependent variable' to indicate that the structure of entrepreneurial networks depends on independent variables such as firm strategy, stage of development, and dynamic information needs of the founders.

The costs and benefits of network relations may largely depend on the state of development of a company. Thus, it would come as no surprise if entrepreneurs change their network utilization patterns over time. One example is the reputation of a 'famous angel' in a start-up's board of directors. This person's value is probably highest in the early phases of market entry when the start-up tries to find the first customers and recruits the first employees. The importance of the famous angel for the start-up diminishes over time, because the company grows, becomes better known to market partners, and builds a reputation of its own.

So far, there have been very few studies on the dynamics of network relations or dynamic versions of the entrepreneurial network success hypothesis (Larson 1992, Larson and Starr 1993, Hite 2000). Larson (1992) as well as Larson and Starr (1993) have presented a dynamic model for the development of entrepreneurial networks. The authors distinguish between three consecutive phases of entrepreneurial networking: focusing on the essential dyads, converting dyadic ties to socio-economic exchanges, and layering the exchanges with multiple exchange processes. The whole process of network building is assumed to involve the exploration, screening, and selective use of network dyads to match the business definition of the emerging firm. The entrepreneur uses existing ties to create new, 'opportunistic ties' (Larson and Starr 1993: 6) intended to foster the success of the start-up. By increasing the number of business ties in the personal network, by building trust with business partners, and by layering existing exchange relations with additional business functions and contact partners, the entrepreneur develops her personal network to an organizational network of her firm. The interesting feature of the model developed by Larson and Starr is that it postulates the dynamic transformation process from personal entrepreneurial networks to organizational networks of start-ups firms for the first time. Thus, it allows entrepreneurship theory to be integrated with existing theories on the networks of firms and their dynamics (Thorelli 1986, Albach 1994).

Witt (1999) has formulated a chaos theoretical model for the development of a firm's centrality in the information network. The model assumes that a fixed proportion of all network ties cannot be kept over time, i.e. that a firm permanently loosens ties, but that it is also able to constantly acquire new contacts. Depending on the loss rate, the acquisition success, and the number of potential network partners, the model derives forecasts for the dynamic development of a firm's position in its information network. For certain parameter constellations, chaotic developments, i.e. very quick and strong changes in a company's network structure are shown to be possible and have also been reported in empirical studies on the networks of transformation firms in East Germany after 1990 (Albach 1994).

Lechner (2001) as well as Lechner and Dowling (2003) have developed a dynamic model of network utilization. They analyse a sample of German software companies empirically and distinguish between four different stages of firm development. The main finding of the study is that entrepreneurs use different types of network ties in each stage of development, i.e. the value of individual network ties for an entrepreneur depends on the current status of her start-up. 'Strong' social network ties and so-called 'reputation networks' matter most in the early phases of the foundation process, they become less important in later phases when the start-up has been able to build a reputation of its own. Marketing networks and so-called 'co-opetition networks' are the main issues of network management during the second stage, technology partnering becomes the focus in the third stage when firms have already built larger networks. In general, 'weak' business ties increase in importance over time, because they serve as 'knowledge and reciprocity networks'. A shortcoming of the 4-stage model by Lechner and Dowling (2003) is that the authors give no clear and measurable definitions for the different network types and do not indicate whether their results are specific for the selected industry or could be the basis of a more general theory of network dynamics.

Dealing with network dynamics, qualitative and inductive research as suggested by Larson (1992) as well as by Hoang and Antoncic (2003) is not the only option. It may not even be the preferable research method due to the lack of generalizability and the descriptive rather than predictive nature of case study research. Equally fruitful and methodologically more rigorous are panel studies with larger sample sizes that allow for quantitative comparative static analyses as suggested by Johannisson (2000). Despite the data collection problems which can be substantial if the sample is large and the time period of repeated enquiries long, the big advantage of panel studies with larger samples is the ability to track changes in networking activities, network structure, and benefits derived from the network over time and thus to draw general conclusions on network dynamics. Clearly, here is ample opportunity for further research.

5.2 Integration of sociological and economic network theories

Sociological approaches to explain economic transactions between network partners focus on social capital (Hansen 2000), social competences (Baron and Markman 2000), mutual trust (Gulati 1995), moral standards (Granovetter 1985), and altruism (Uzzi 1997). Thus, this line of research predicts that entrepreneurs can get access to cheap or otherwise inaccessible resources because their network partners do behave as *homo oeconomici*, i.e. they do not maximize their own economic utility. Non-economic motives are said to dominate exchange relations in networks. Even if this was generally true, the question that remains unanswered is what entrepreneurs need to do for their network partners, or, in other words, which non-economic benefits they contribute to the network to induce mutual trust, altruism, and social capital with their partners.

There is little doubt that social relations exist in which one person supports the other for reasons of pure idealism or altruism, i.e. without expecting anything in exchange.[5] In general, one must expect the principle of reciprocity (Larson 1992, Fehr and Gächter 1998) that was shown to influence economic transactions largely, to also hold for social relations in networks. Thus, it will only be possible for an entrepreneur to obtain cheap or otherwise inaccessible resources via network ties if she herself also offers services and information at prices below market prices to her network partners. In the long run, the ties need to be symmetrical and based on contributions of equal (subjective) value. Entrepreneurs trying to utilize their network ties opportunistically without reciprocal offerings are bound to fail. 'If the strong assumptions of trust and co-operation are exploited in embedded ties, vendettas and endless feuds can arise' (Uzzi 1997: 59). The fact that it is difficult to measure the costs of reciprocal economic and social services in entrepreneurial networks (Starr and McMillan 1990: 90) is no justification to ignore them.

From an economist's perspective, the most important shortcoming of existing sociological versions of entrepreneurial network theory is that they are 'oversocialized'. One certainly cannot analyse economic transactions without referring to psychological and sociological phenomena, but we cannot explain them without referring to economic motivations either. 'Actors are oversocialized when portrayed as governed exclusively by values and norms and undersocialized when portrayed as isolated, rational economic units' (Larson 1992: 97). To our best understanding, all transaction in networks, no matter how strongly they are embedded in social relations, are based on the decisions taken by utility-maximizing individuals. Social and even altruistic motives can (and will) be part of individual persons' utility, but economic motives always remain present.

An example from the literature clearly shows the mistakes researchers can make when they base their analysis of entrepreneurial networks on social embeddedness only. Uzzi (1999) concludes from an empirical study of entrepreneurs and their relations to banks that socially embedded ties lead to lower interest rates to be paid on bank loans than arm's length relations. The author measures the degree of embeddedness by the duration of the relation between bank and entrepreneur and the scope of exchange relations between the two partners. His recommendation for prospective entrepreneurs is to establish a minimum of tight, socially embedded ties to banks (more precisely: bank employees in charge of loan decisions) to realize lower costs of capital. From an economic perspective, another explanation for Uzzi's (1999) empirical findings is much more convincing: banks will simply be able to offer lower interest rates on bank loans if they have known a customer for a longer time and if that customer buys more than one product from the bank. Long customer relations reduce a bank's cost to check credit worthiness and to obtain information on the customer. They serve to reduce transaction costs. Multiple customer relations allow for cross-product calculations. Both effects can easily explain interest rates to be lower in what Uzzi (1999) depicts as 'embedded ties'. Thus, it can be concluded that the (rather sociological) network approach and the (rather transaction cost oriented) economic approach differ in their basic assumptions (Johansson and Matsson 1987), but still can be combined fruitfully in future theoretical work on entrepreneurial networking.

The suggestions for further research that have been made in this section of the paper all add to the need to empirically test the hypotheses which have been formulated in the preceding section.

Acknowledgements

In preparing this paper I benefited greatly from comments by Horst Albach, Klaus Brockhoff, and Peter-J. Jost. German Brachtendorf, Verena Rode, and Andreas Schroeter helped in reviewing the existing literature on the subject. Earlier versions of this paper were presented in February 2002 at the Annual Conference of the Commission on Organization Theory of the German Association of University Professors in Management in Lüneburg; in July 2002 as my habilitation lecture at Humboldt-University in Berlin; and in February 2003 at the first Interdisciplinary European Conference on Entrepreneurship Research (IECER) in Regensburg. The author would like to thank two anonymous referees and the editor of this journal for their substantial advice and guidance in revising this paper. All remaining errors and omissions are my own responsibility.

Notes

1. McGee, Dowling and Megginson (1995) analyse a sample of 210 stock-listed growth companies based on information given in IPO documents. They measure networking activities of these corporations with a dichotomous variable, the existence (or non-existence) of co-operations ranging from informal agreements to contractually fixed joint ventures in the areas of marketing, R&D, and production. The study chooses the compound growth rate of sales over a 3-year period as the proxy for corporate success.
2. Baron and Markman (2000) have also put forward this argument and developed a theory of social skills of entrepreneurs.
3. The study of Brüderl and Preisendörfer (1998) is the only existing empirical study of the network success hypothesis that goes beyond measuring networking activities and structural network measures and focuses on the amount of network support entrepreneurs received. Unfortunately, their paper does not make an attempt to compare the amount of support received, i.e. the benefits from the network, with the costs of building, maintaining, and extending it.
4. An exception is Johannisson (2000) who investigates the dynamics of serial entrepreneurs and presents the entrepreneurial career as a set of interlocking ventures embedded in the personal network of the entrepreneur (which is assumed to be constant).
5. This may be most plausible for the relation between parents and their children. Altruism may also be the dominant motive of some business angels supporting young entrepreneurs, but certainly not for all of them.
6. Johansson and Mattsson (1987) have compared the network approach with the transaction cost approach in all details. They show that the network approach in its general form views firms as social units and is closer to social exchange theory than to neoclassical economic theory. The authors also point out that opportunistic behaviour, a central assumption in transaction cost approaches, is replaced by mutual trust and personal relations in network approaches.

References

Albach, H. 1994 *The Transformation of Firms and Markets. A Network Approach to Economic Transformation Processes in East Germany* (Uppsala: Uppsala University Press).

Albach, H. and Kless, H. 1982 Personelle Verflechtungen bei deutschen Industrieaktiengesellschaften, *Zeitschrift für Betriebswirtschaft*, 52: 959–977.

Aldrich, H. E. and Reese, P. R. 1993 Does networking pay off? A panel study of entrepreneurs in the research triangle, in Churchill, N. C., Birley, S., Doutriaux, J., Gatewood, E. J., Hoy, F. S. and Wetzel, W. E. Jr (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 325–339.

Aldrich, H. E. and Zimmer, C. 1986 Entrepreneurship through social networks, in Sexton, D. L. and Smilor, R. (eds), *The Art and Science of Entrepreneurship* (Cambridge, MA: Ballinger) pp. 3–23.

Aldrich, H. E., Rosen, B. and Woodward, W. 1987 The impact of social networks on business foundings and profit: a longitudinal study, in Churchill, N. C., Homaday, J. A., Kirchhoff, B. A., Krasner, O. J. and Vesper, K. H. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 154–168.

Aldrich, H. E., Reese, P., Dubini, P., Rosen, B. and Woodward, B. 1989 Women on the verge of a breakthrough: networking between entrepreneurs in the United States and Italy, *Entrepreneurship & Regional Development*, 1: 339–356.

Baron, R. A. and Markman, G. D. 2000 Beyond social capital: how social skills can enhance entrepreneurs' success, *Academy of Management Executive*, 14: 106–116.

Bavelas, A. 1948 A mathematical model for group structure, *Applied Anthropology*, 7: 16–30.

Bayer, K. 1991 The impact of using consultants during venture formation on venture performance, in Churchill, N. C., Bygrave, W. D., Covin, J. G., Sexton, D. L., Slevin, D. P., Vesper, K. H. and Wetzel, W. E. Jr. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 291–305.

Birley, S. 1985 The role of networks in the entrepreneurial process, *Journal of Business Venturing*, 1: 107–117.

Birley, S., Cromie, S. and Myers, A. 1991 Entrepreneurial networks: their emergence in Ireland and overseas, *International Small Business Journal*, 9: 56–73.

Bouwen, R. and Steyaert, C. 1990 Construing organizational texture in young entrepreneurial firms, *Journal of Management Studies*, 26: 637–649.

Brüderl, J. and Preisendörfer, P. 1998 Network support and the success of newly founded businesses, *Small Business Economics*, 10: 213–225.

Brush, C. G. and Vanderwerf, P. A. 1992 A comparison of methods and sources for obtaining estimates on new venture performance, *Journal of Business Venturing*, 7: 157–170.

Burt, R. S. 1992 *Structural Holes: The Social Structure of Competition* (Cambridge, MA: HBS Press).

Chandler, G. N. and Hanks, S. H. 1993 Measuring the performance of emerging businesses: a validation study, *Journal of Business Venturing*, 8: 391–408.

Chell, E. and Baines, S. 1998 Does gender affect business performance? A study of microbusinesses in business services in the UK, *Entrepreneurship & Regional Development*, 10: 117–135.

Chell, E. and Baines, S. 2000 Networking, entrepreneurship and microbusiness behaviour, *Entrepreneurship & Regional Development*, 12: 195–215.

Chicha, J. 1980 *Les strategies des PME et leur adaption au changement*, Working Paper No. 80-05, GREPME, Trois-Rivières University, Trois-Rivières.

Cohen, W. M. and Levinthal, D. A. 1990 Absorptive capacity: a new perspective on learning and innovation, *Administrative Science Quarterly*, 35: 128–152.

Cooper, A. C., Folta, T. and Woo, C. Y. 1991 Information acquisition and performance by start-up firms, in Churchill, N. C., Bygrave, W. D., Covin, J. G., Sexton, D. L., Slevin, D. P., Vesper, K. H. and Wetzel, W. E. Jr. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 276–290.

Daft, R. L. and Lengel, R. H. 1986 Organizational information requirements, media richness and structural design, *Management Science*, 32: 554–571.

Davidsson, P. 1989 Entrepreneurship and after? A study of growth willingness in small firms, *Journal of Business Venturing*, 4: 211–226.

Drakopoulou Dodd, S. D. and Patra, E. 2002 National differences in entrepreneurial networking, *Entrepreneurship & Regional Development*, 14: 117–134.

Dubini, P. and Aldrich, H. E. 1991 Personal and extended networks are central to the entrepreneurial process, *Journal of Business Venturing*, 6: 305–313.

Ebers, M. and Grandori, A. 1997 The forms, costs, and development dynamics of inter-organizational networking, in Ebers, M. (ed.), *The Formation of Inter-Organizational Networks* (Oxford: Oxford University Press) pp. 265–286.

Fehr, E. and Gächter, S. 1998 Reciprocity and economics: the economic implications of *Homo Reciprocans*, *European Economic Review*, 42: 845–859.

Freeman, L. C. 1978/79 Centrality in Social Networks, conceptual clarification, *Social Networks*, 1: 215–239.

Granovetter, M. S. 1973 The strength of weak ties, *American Journal of Sociology*, 78: 1360–1380.

Granovetter, M. S. 1985 Economic action and social structure: the problem of embeddedness, *American Journal of Sociology*, 91: 481–510.

Gulati, R. 1995 Does familiarity breed trust? The implications of repeated ties for contractual choice in alliances, *Academy of Management Journal*, 38: 65–112.

Hansen, E. L. 1995 Entrepreneurial networks and new organization growth, *Entrepreneurship Theory & Practice*, 19: 7–19.

Hansen, E. L. 2000 Resource acquisitions as a start-up process: initial stocks of social capital and organizational foundings, in Reynolds, P. D., Autio, E., Brush, C. G., Bygrave, W. D., Manigart, S., Sapienza, H. J. and Shaver, K. G. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 111–120.

Hippel, E. 1994 'Sticky information' and the locus of problem solving: implications for innovation, *Management Science*, 40: 429–439.

Hite, J. M. 2000 Qualities of embedded network ties of emerging entrepreneurial firms, in Reynolds, P. D., Autio, E., Brush, C. G., Bygrave, W. D., Manigart, S., Sapienza, H. J. and Shaver, K. G. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 121–133.

Hoang, H. and Antoncic, B. 2003 Network-based research in entrepreneurship. A critical review, *Journal of Business Venturing*, 18: 165–187.

Jarillo, J. C. 1989 Entrepreneurship and growth: the strategic use of external resources, *Journal of Business Venturing*, 4: 133–147.

Johannisson, B. 1988 Business formation – a network approach, *Scandinavian Journal of Management*, 4: 83–99.

Johannisson, B. 1996 The dynamics of entrepreneurial networks, in Reynolds, P. D., Birley, S., Butler, J. E., Bygrave, W. D., Davidsson, P., Gartner, W. B. and McDougall, P. P. (eds), *Frontiers of Entrepreneurship Research* (Wellesley, MA: Babson College) pp. 253–267.

Johannisson, B. 2000 Networking and entrepreneurial growth, in Sexton, D. L. and Landström, H. (eds), *The Blackwell Handbook of Entrepreneurship* (Oxford: Oxford University Press) pp. 368–386.

Johannisson, B. and Nilsson, A. 1989 Community entrepreneurs networking for local development, *Entreprenurship & Regional Development*, 1: 3–19.

Johansson, J. and Mattson, L.-G. 1987 Interorganizational relations in industrial systems: a network approach compared with the transaction-cost approach, *International Studies of Management & Organization*, 17: 34–48.

Kirchhoff, B. A. 1994 Entrepreneurship and dynamic capitalism, *The Economics of Business Firm Foundation and Growth* (Westport: Quorum Books).

Larson, A. 1992 Network dyads in entrepreneurial settings: a study of the governance of exchange relationships, *Administrative Science Quarterly*, 37: 76–104.

Larson, A. and Starr, J. A. 1993 A network model of organization formation, *Entrepreneurship Theory & Practice*, Winter: 5–15.

Lechner, C. 2001 *The Competitiveness of Firm Networks* (Frankfurt: Peter Lang).

Lechner, C. and Dowling, M. 2003 Firm networks: external relationships as sources for the growth and competitiveness of entrepreneurial firms, *Entrepreneurship & Regional Development*, 15: 1–26.

Lee, D. Y. and Tsang, E. W. 2001 The effects of entrepreneurial personality, background and network activities on venture growth, *Journal of Management Studies*, 38: 583–602.

Lincoln, J. R. 1982 Intra- (and inter-)organizational networks, in Bacharach, S. (ed.), *Research in the Sociology of Organizations* (Greenwich, CT: JAI Press) pp. 1–38.

Littunen, H. 2000 Networks and local environmental characteristics in the survival of new firms, *Small Business Economics*, 15: 59–71.

McGee, J. E., Dowling, M. J. and Megginson, W. L. 1995 Cooperative strategy and new venture performance: the role of business strategy and management experience, *Strategic Management Journal*, 16: 565–580.

Niemeijer, R. 1973 Some applications of the notion of density to network analysis, in Boissevain, J. and Mitchell, C. (eds), *Network Analysis – Studies in Human Interaction* (Paris: Mouton) pp. 45–64.

Nieminen, J. 1974 On the centrality in a graph, *Scandinavian Journal of Psychology*, 15: 332–336.

Ostgaard, T. A. and Birley, S. 1996 New venture growth and personal networks, *Journal of Business Research*, 36: 37–50.

Sahlmann, W. A. 1990 The structure and governance of venture-capital organizations, *Journal of Financial Economics*, 27: 473–521.

Salancik, G. R. 1995 WANTED: a good network theory of organization, *Administrative Science Quarterly*, 40: 345–349.

Smith, N. R. 1967 *The Entrepreneur and His Firm: The Relationship between Type of Man and Type of Company* (East Lansing, MI: Michigan State University Press).

Starr, J. A. and Macmillan, I. C. 1990 Resource co-optation via social contracting: resource acquisition strategies for new ventures, *Strategic Management Journal*, 11: 79–92.

Thorelli, H. 1986 Networks: between markets and hierarchies, *Strategic Management Journal*, 7: 37–51.

Uzzi, B. 1997 Social structure and competition in interfirm networks: the paradox of embeddedness, *Administrative Science Quarterly*, 42: 35–67.

Uzzi, B. 1999 Embeddedness in the making of financial capital: how social relations and networks benefit firms seeking finance, *American Sociological Review*, 64: 481–505.

Walker, G. 1988 Network analysis for cooperative interfirm relationships, in Contractor, F. and Lorange, P. (eds), *Comparative Strategies in International Business* (Lexington: Rowman & Littlefield) pp. 227–240.

Witt, P. 1999 Information networks of small and medium-sized enterprises, *Journal of Enterprising Culture*, 7: 213–231.

Witt, P. and Rosenkranz, S. 2002 Netzwerkbildung und Gründungserfolg, *Zeitschrift für Betriebswirtschaft*, 72 (Special Issue 5): 85–105.